Initiation à la recherche

PR422 – PR423

SUJETS ETAT DE L'ART

2026-2027

Table des matières

AI-AES: AI-Assisted Cryptanalysis of Reduced-Round AES 3

CLF4Drone: Control Lyapunov Functions for Quadcopter Stabilization 4

CTD-NL : Comment concevoir une commande tolérante aux défauts d'actionneurs, de capteurs pour les systèmes non linéaires incertains ? 5

DCMWSAN: Distributed Communication in Mobile Wireless Sensor/Actuator Networks 6

DRIFT-AG: Detection and Compensation of Drift in Low-Cost Electrochemical Soil Sensors for Embedded Precision Agriculture 7

Fraudulent: Fraudulent research in computer and electronic science 8

HDL_Proof: formally proving circuits with proof assistants 9

Hotpatch: Hot patching of an application 10

IASPEC : un LLM avec ses propres données 11

LLM-CRYPT: Can LLMs Perform Cryptanalysis? 12

LST: Latent Space Translation 13

NeuroLyap : Comment prouver la stabilité d'une commande non linéaire à base d'approximation par réseaux de neurones en présence d'incertitudes et de perturbations ? 14

OPTSECCOMP: Optimisations et sécurité dans les compilateurs 15

PLANARF: PLANar Antenna and RF Rectification 16

Qcar_Control: Model Predictive Control and Differential Flatness-Based Approaches for the QCar: Control and Collision Avoidance Strategies 17

RECONF-DET: Reconfigurable attack detectors in integrated circuits 18

SML: Similarity and Metric Learning 19

TDCFID: Time to Digital Converter for Fault Injection Detection 20

AI-AES: AI-Assisted Cryptanalysis of Reduced-Round AES

Supervisor(s): Hoa Nguyen

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Research problem: How did an autonomous AI system contribute to improving cryptanalysis of reduced-round AES, and how can such an AI-generated cryptanalytic result be understood, verified, and assessed by human researchers?

Short description: AES is one of the most widely used encryption standards. Like many cryptographic algorithms, it has been studied through simplified versions in which the number of internal processing steps, called rounds, is reduced. These simplified versions are used by researchers to understand the security margin of the full algorithm; they do not represent a practical break of the AES standard used in real applications.

In 2026, Nasr and Carlini reported an improved attack on a reduced 7-round version of AES-128. The work was motivated by an autonomous AI system that identified a new mathematical shortcut, called the "Möbius Bridge", which improved a previous academic attack. This result offers a concrete case study of how AI may assist cryptanalysts in finding and testing new ideas.

The objective of this project is to read and analyze this work in its scientific context. Students will first introduce the basic principles of AES and reduced-round cryptanalysis. They will then study the family of attacks on 7-round AES, the idea behind the improvement proposed by Nasr and Carlini, and the methods used to validate the result. Finally, they will discuss the respective roles of AI and human researchers, the limits of the result, and why an improvement against 7-round AES does not imply a weakness in full AES-128.

Reference(s):

[1] Anthropic. Discovering Cryptographic Weaknesses with Claude. Anthropic Research, 2026. Available: https://www.anthropic.com/research/discovering-cryptographic-weaknesses

[2] Nasr, Milad, and Nicholas Carlini. "Cryptanalysis of 7-Round AES via the Algebraic Structure of its S-box." https://www-cdn.anthropic.com/c88771e1bf5ee8885349eed05e5484c0e5f7e02b/aes_mobius_bridge.pdf

CLF4Drone: Control Lyapunov Functions for Quadcopter Stabilization

Supervisor: Ionela Prodan (ionela.prodan@lcis.grenoble-inp.fr)

Title: Control Lyapunov Functions for Quadcopter Stabilization

Research problem: What are Control Lyapunov Functions (CLFs) and how they are used to guarantee stability and performance in nonlinear systems such as quadcopters and how can they be integrated into Model Predictive Control (MPC) frameworks to achieve real-time feasible control?



Short description: A Control Lyapunov Function (CLF) is a special kind of scalar function used to design stabilizing controllers for nonlinear systems. It extends the idea of a Lyapunov function (used to analyse stability) by incorporating control inputs, so that it can be used constructively to design a control law that guarantees stability. Students will focus on understanding: i) the mathematical definition and properties of Control Lyapunov Functions; ii) the relationship between CLFs and MPC, and how they ensure system stability; iii) the design and use of CLFs for multicopter (quadcopter) stabilization; iv) the methods proposed in the papers to reduce real-time computational complexity and improve experimental feasibility.

Expected outcomes: The students should read at least 3 recent papers on this topic and concentrate on: i) a structured literature review summarizing the theoretical foundations of CLFs, ii) the relationship between MPC and CLF in the literature; iii) a conceptual explanation of how these methods are applied to multicopter control and what models are used; iv) a short summary diagram or flowchart illustrating how CLF-based MPC stabilizes a Crazyflie quadcopter; v) a discussion of the trade-offs between theoretical guarantees and real-time implementation constraints.

References:

[1] HT Do, F Blanchini, S Miani, I Prodan (2024): LP-generated Control Lyapunov Functions with application to multicopter control. IEEE Transactions on Control Systems Technology 32 (6), 2090-2101

[2] HT Do, F Blanchini, S Miani, I Prodan (2025): Reducing real-time complexity via sub-control Lyapunov functions: from theory to experiments. Automatica 183, 112592

CTD-NL : Comment concevoir une commande tolérante aux défauts d'actionneurs, de capteurs pour les systèmes non linéaires incertains ?

Supervisor(s): Islam DAOUDI

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Research problem:

Comment garantir la stabilité et préserver les performances des systèmes non linéaires en présence d'incertitudes, de perturbations et de différents types de défaillances affectant les actionneurs, les capteurs ? Quelles stratégies de commande tolérante aux défauts (*Fault-Tolerant Control*, FTC) permettent de faire face à ces défaillances, notamment lorsqu'elles surviennent simultanément, et quelles garanties théoriques de stabilité offrent-elles ?

Short description:

Les stratégies FTC se divisent principalement en trois catégories : les approches passives, fondées sur des lois de commande robustes, les approches actives, reposant sur la détection et la compensation des défauts, et les approches adaptatives, qui ajustent les paramètres de commande en fonction des défaillances.

L'objectif de ce projet est de réaliser un état de l'art des méthodes de commande tolérante aux défauts des systèmes non linéaires incertains, en s'intéressant particulièrement aux aspects suivants :

1. **Classification et modélisation des défauts :** étudier les différents types de défaillances et leurs représentations mathématiques.
2. **Stratégies FTC :** analyser les approches passives, actives et adaptatives, ainsi que leurs principes de fonctionnement et leurs domaines d'application.
3. **Détection, estimation et compensation :** examiner les méthodes permettant d'identifier, d'estimer et de compenser les défauts d'actionneurs, de capteurs, y compris en cas de défaillances simultanées.

Reference(s):

[1] Daoudi, Islem, et al. "Sensor and actuator fault-tolerant fuzzy adaptive control for fractional-order strict feedback nonlinear systems subject to input saturation." *IEEE Access* 13 (2025): 91706-91723.

[2] Daoudi, I., Bounemeur, A., Rabahi, Z., & Chemachema, M. (2026). Observer-Driven Event-Triggered Adaptive Fault-Tolerant Control of Fractional-Order Nonlinear Systems under Input Saturation. *Iranian Journal of Science and Technology, Transactions of Electrical Engineering*, 1-18

[3] DAOUDI, Islam, RABAHI, Zoubir, et CHEMACHEMA, Mohamed. Finite-time fractional-order fault-tolerant neural-network adaptive control for fractional-order strict-feedback nonlinear systems subject to input saturation. *Transactions of the Institute of Measurement and Control*, 2026.

DCMWSAN: Distributed Communication in Mobile Wireless Sensor/Actuator Networks

Supervisors: Louis Duvivier, Clément Raïevsky et Laurent Lefèvre

Supervisor's emails: nom.prenom@lcis.grenoble-inp.fr

Research problem: develop a distributed algorithm to build the ad hoc communication network topology between mobile agents. A set of mobile sensors/actuators (MWSAN), communicating via a wireless ad hoc network (with a dynamic structure), must approach predefined targets while maintaining network connectivity.

Short description:

The development of autonomous robots and drones raises the issue of their ability to operate independently in groups. There are numerous applications for wireless sensor networks: in material handling, surveillance or rescue operations in hazardous environments, agricultural or industrial production, etc. In addition to being equipped with sensors, mobile robots may have actuators and interact with their environment, which greatly expands the range of possible applications.

For the group of agents to operate autonomously, it must be able to manage not only each agent individually but also their collaboration within a network. It is therefore necessary to dynamically deploy and maintain the communication network among the agents, ensuring that the network structure is well-suited to the tasks at hand and adhering to constraints such as communication range. We are interested in unconstrained communication network structures capable of adapting to the tasks at hand. Agents may be called upon to play different roles within the system depending on their positioning, but these roles are not predetermined.

The communication network structure must be adapted in a distributed manner, that is, without centralized coordination of all agents. This is a significant constraint for reasons of both scalability and system robustness. Centralized management involves an excessive amount of communication and also weakens the system by introducing single points whose failure is critical to the entire group.

Reference(s):

[1] M. M. Zavlanos and G. J. Pappas, "Distributed connectivity control of mobile networks," IEEE Transactions on Robotics, vol. 24, no. 6, pp. 1416–1428, 2008.

[2] A. Gusrialdi, "Connectivity-preserving distributed algorithms for removing links in directed networks," Network Science, vol. 10, no. 3, pp. 215–233, 2022

[3] D. B. Venkateswaran, Z. Qu, and A. Gusrialdi, "A distributed method for detecting critical edges and increasing edge connectivity in undirected networks," in 2024 IEEE 63rd Conference on Decision and Control (CDC). IEEE, 2024, pp. 6951–6956.

DRIFT-AG: Detection and Compensation of Drift in Low-Cost Electrochemical Soil Sensors for Embedded Precision Agriculture

Supervisor(s): Celine Jeanne

Supervisor's emails: celine.jeanne@esisar.grenoble-inp.fr

Research problem: Detection and compensation of drift in low-cost electrochemical sensors deployed in agricultural field conditions

Short description: Low-cost electrochemical sensors measuring soil parameters (pH, electrical conductivity, nitrogen, phosphorus, potassium) deployed in precision agriculture are subject to progressive measurement drift caused by electrode ageing, seasonal climate variations, and fouling under real field conditions. If undetected, this drift can lead to erroneous agronomic recommendations based on biased data. The literature on concept drift detection in IoT data streams proposes several approaches — adaptive windowing statistical methods (ADWIN, Page-Hinkley), machine learning-based methods, and empirical compensation models. The objective of this literature review is to identify and compare existing methods for detecting and compensating drift in low-cost electrochemical sensors deployed outdoors over extended periods (several months to one year), taking into account the specific constraints of a resource-limited embedded system (ESP32 microcontroller) and the difficulty of distinguishing instrumental drift from normal seasonal variation.

Reference(s):

[1] Song et al. — *A new paradigm of reliable sensing with field-deployed electrochemical sensors integrating data redundancy and source credibility* — Scientific Reports, Vol. 13, Article 3101, February 2023. DOI: 10.1038/s41598-022-25920-w

[2] R*eal Time Drift Detection and Adaptation Using Hybrid ADWIN in Agricultural Environmental Monitoring* — International Journal of Electrical and Electronics Engineering & Telecommunications (IJEETC), Vol. 14, No. 5, 2025

Fraudulent: Fraudulent research in computer and electronic science

Supervisor(s): **Jean-Luc Koning**

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Research problem: **Is computer and electronic science at risk of fraudulent research?**

Short description: The COVID-19 pandemic triggered an unprecedented surge in fake, fraudulent, and deeply flawed research papers, a phenomenon frequently referred to by experts as a "paperdemic". Driven by rushed peer-review processes, the professional pressure to "publish or perish," and a global demand for immediate answers, malicious actors and desperate researchers successfully bypassed traditional scientific safeguards.

The most notorious breakdown in scientific publishing occurred when The Lancet and the New England Journal of Medicine (NEJM) were forced to retract major studies after discovering they relied on entirely fabricated, suspect international patient data provided by a little-known company called Surgisphere. [3]

Major Drivers of Fake COVID-19 Papers

1. Accelerated Peer Review: Journals slashed review times to days, leading to gaps in verifying methodology or identifying counterfeited data.
2. Paper Mills: Commercially motivated "paper mills" ramped up production, selling completely fabricated, template-based biomedical manuscripts to researchers needing quick publications.
3. Ideological & Political Agendas: Fraudulent data was frequently generated to rapidly support or debunk highly politicized treatments, such as hydroxychloroquine.

Does this problem also affect scientific fields such as computer science or electronics [2]? To what extent [1]?

Reference(s):

[1] Fake papers are contaminating the world's scientific literature, fueling a corrupt industry and slowing legitimate lifesaving medical research. The Conversation, 29 janvier 2025, Cyril Labbé & Guillaume Cabanac https://theconversation.com/fake-papers-are-contaminating-the-worlds-scientific-literature-fueling-a-corrupt-industry-and-slowing-legitimate-lifesaving-medical-research-246224

[2] Fraudulent research is 'destroying trust in science'. Matthew Ward Agius. 08/10/2025. DW. https://www.dw.com/en/fake-studies-on-the-rise-destroying-trust-in-science/a-73533918

[3] Biased, wrong and counterfeited evidences published during the COVID-19 pandemic, a systematic review of retracted COVID-19 papers, Quality & Quantity, Springer, Volume 57, pages 4881–4913 (2023). Angelo Capodici, Aurelia Salussolia, Francesco Sanmarchi, Davide Gori & Davide Golinelli. https://link.springer.com/article/10.1007/s11135-022-01587-3

HDL_Proof: formally proving circuits with proof assistants

Supervisor(s): KIEFFER Yann

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Research problem: Explore how proof assistants have been used for formally verifying the function of a digital design.

Short description: Errors in software implementations are a problem, which many times translate into increased implementation cost. Errors in hardware digital design are much worse: they may translate into an unusable product. This is why there is a strong focus on formal methods for digital designs.

Although there are many such methods, one class of methods stands apart, that which makes use of so-called *proof assistants*. Proof assistants are software that are designed to help the user formally verify a mathematical proof. They have also been used to verify software. These tools are somewhat harder to use; but they are limited only by what one can prove using mathematics, so they are quite powerful.

In this work, you are to explore how proof assistants have been put to use on the task of verifying digital designs. Example works are cited below, an older one for VHDL [1], and a more recent one for Verilog [2].

Reference(s):

[1] Van Tassel, John Peter. *Femto-VHDL: The Semantics of a Subset of VHDL and its Embedding in the HOL Proof Assistant*. No. UCAM-CL-TR-317. University of Cambridge, Computer Laboratory, 1993.

[2] Loow, Andreas, and Magnus O. Myreen. "A Proof-Producing Translator for Verilog Development in HOL." *2019 IEEE/ACM 7th International Conference on Formal Methods in Software Engineering (FormaliSE)*. 2019.

Hotpatch: Hot patching of an application

Supervisor(s): Quentin Giorgi

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Research problem: Hot patching of an application

Short description:

Hot patching consists of modifying an application while it is running. Normally, applying a patch to an application requires restarting it, which results in the loss of its current state, such as its current execution point or the values of its variables. One proposed solution is therefore to apply patches without having to reset the application's state.

Reference(s):

[1] Ramaswamy A., Bratus S., Smith S. W. & Locasto M. E. (2010, February). Katana: A hot-patching framework for elf executables. In 2010 International Conference on Availability, Reliability and Security (pp. 507-512). IEEE.

[2] Linux foundation. Linux Kernel Documentation - Livepatch. Version 6.12. https://www.kernel.org/doc/html/v6.12/livepatch/livepatch.html

[3] Niesler C., Surminski S. & Davi L. (2021, February). HERA: Hotpatching of Embedded Real-time Applications. In NDSS.

IASPEC : un LLM avec ses propres données

Supervisor(s): Emmanuel BRUN

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Research problem: Utilisation d'un moteur d'IA générative sur des données privées

Short description:

IL existe actuellement de nombreux moteurs d'IA générative (LLM) capables répondre de nombreuses questions, pourvu que la réponse soit accessible publiquement. Il existe également différentes méthodes permettant de "fournir" à un LLM un ensemble de documents, et ensuite le LLM est capable de produire une réponse à une question en se basant sur les documents fournis.

L'objectif de ce travail est de réaliser un état de l'art des outils disponibles pour résoudre cette problèmatique. L'étude identifiera et classifiera les approches existantes pour résoudre ce type de problèmes.

LLM-CRYPT: Can LLMs Perform Cryptanalysis?

Supervisor(s): Hoa Nguyen

Supervisor's emails: phuong-hoa.nguyen@lcis.grenoble-inp.fr

Research problem: Can Large Language Models help analyze the security of cryptographic systems? What can they do well, and where are their current limitations?

Short description: Cryptography protects information in everyday applications such as online payments, messaging, cloud storage, and connected devices. Cryptanalysis is the study of methods used to evaluate whether cryptographic systems contain weaknesses.

Recent progress in Large Language Models (LLMs) raises a new question: can these AI systems help experts analyze cryptographic systems? This project will review recent work on the use of LLMs for cryptanalysis, starting from the **CryptanalysisBench** benchmark.

Students will study the types of problems given to LLMs, the methods used to evaluate their answers, and the ways researchers check whether a proposed attack or weakness is actually correct. They will compare tasks such as explaining a cryptographic construction, reproducing a known weakness, writing experimental code, and suggesting new ideas.

The review will also discuss the limits of current AI systems, including incorrect answers, incomplete code, high computational cost, difficulties in reproducing results, and the difference between toy examples used for research and cryptographic systems used in real applications.

Reference(s):

[1] Fluri, Lukas, Avital Shafran, Nicholas Carlini, Matthew Jagielski, Milad Nasr, Orr Dunkelman, Eyal Ronen, and Florian Tramèr. "CryptanalysisBench: Can LLMs do Cryptanalysis?" *arXiv preprint arXiv:2607.18538* (2026).

LST: Latent Space Translation

Supervisor(s): Louis Morge-Rollet, Clément Raïevsky

Supervisor's emails: louis.morge-rollet@grenoble-inp.fr, clement.raievsky@univ-grenoble-alpes.fr

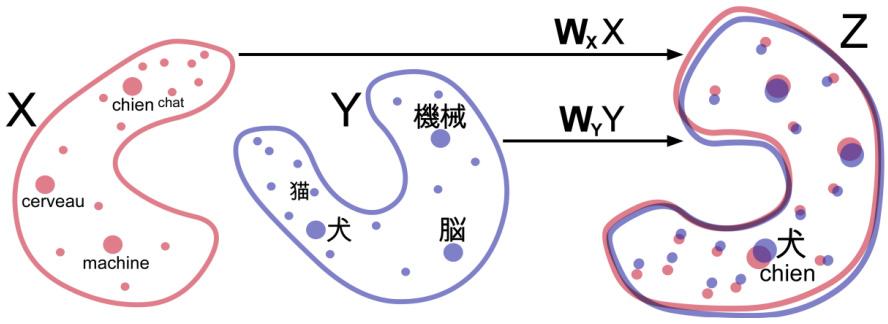
Research problem: What are the methods for latent space translation (for stream of data)?

Short description:

Deep learning (DL), is a part of Machine Learning (ML), and consists using (deep) neural networks, a bio-inspired mathematical model of the brain, to solve complex problems (image classification, text generation, …). A latent space, or embedding space, is a vector space used to represents data. It can be learned using auto-encoders, a particular type of neural networks, and used to represent an input data (image, sound, …) in an abstract way, as a vector.

Generally, a latent space is learned only for a particular type of data. However, relatively recently, researchers tried to find a way to perform latent space translation, i.e. to pass from one latent space to another latent space. This kind of approach have several applications such as text-to-image generation (generation of an image from a textual description) or image-to-text (generation of a textual description from an image). One of the first work on this task was Contrastive Language-Image Pre-training (CLIP), developed by a OpenAI research team, and consisting to align latent spaces of images and texts through contrastive learning [1]. However, Global Latent Workspace (GLT) extends this approach to other kind of data (sound, …) [2].

The study will identify the different methods proposed in the literature of latent space translation. Particularly, after identifying the principal methods, the student can focus on methods working on stream of data (audio, video, …).



Exemple d'alignement sémantique entre l'espace latent des mots Français (X) et l'espace latent des mots Chinois (Y) pour un cas d'application à la traduction Français-Chinois [2]

Reference(s):

[1] A. Radford et al., Learning Transferable Visual Models From Natural Language Supervision, Proceedings of Machine Learning Research, 2021.

[2] R. VanRulen et al., Deep Learning and the Global Workspace Theory, Trends in Neurosciences, 2021.

NeuroLyap : Comment prouver la stabilité d'une commande non linéaire à base d'approximation par réseaux de neurones en présence d'incertitudes et de perturbations ?

Supervisor(s): Islam DAOUDI

Supervisor's emails: daoudiislem4@gmail.com

Research problem:

Comment garantir et prouver la stabilité d'un système de commande non linéaire dont les dynamiques inconnues sont approximées par un réseau de neurones /logique flou ? Quelles méthodes d'analyse permettent de prendre en compte l'erreur d'approximation, les incertitudes paramétriques et les perturbations externes, et quel type de stabilité peut-on réellement garantir ?

Short description:

La commande des systèmes non linéaires (formes de Brunovsky, *strict-feedback* et *pure-feedback*) nécessite généralement une connaissance précise du modèle, rarement disponible en pratique. Pour pallier cette difficulté, les réseaux de neurones (RBF, perceptrons multicouches) ou bien les logiques floues sont utilisées pour approximer les dynamiques inconnues, avec des paramètres ajustés en ligne par des lois d'adaptation, Cependant, les erreurs d'approximation, les incertitudes et les perturbations externes compliquent l'analyse de stabilité. La stabilité asymptotique est alors rarement garantie, et les résultats reposent le plus souvent sur la stabilité pratique, notamment la bornitude, établie à l'aide de la théorie de Lyapunov.

L'objectif de ce projet est de réaliser un état de l'art des méthodes de démonstration de la stabilité des commandes non linéaires fondées sur les réseaux de neurones, en abordant :

1. Les hypothèses liées à l'approximation neuronale et aux erreurs résiduelles.
2. Les outils d'analyse de stabilité (Lyapunov, et stabilité semi-globale).
3. Les mécanismes de robustification des lois d'adaptation
4. Les garanties de stabilité obtenues et leurs limites, notamment en matière de convergence et de réglage des gains.

Reference(s):

[1] Rabahi, Z., Daoudi, I., & Chemachema, M. (2024). Neural network-based output-feedback adaptive control for a class of uncertain strict feedback fractional-order nonlinear systems subject to input saturation. *International Journal of Adaptive Control and Signal Processing*, *38*(5), 1690-1709

[2] CHEMACHEMA, Mohamed. Output feedback direct adaptive neural network control for uncertain SISO nonlinear systems using a fuzzy estimator of the control error. *Neural networks*, 2012, vol. 36, p. 25-34.

OPTSECCOMP: Optimisations et sécurité dans les compilateurs

Supervisor(s): Christophe Deleuze

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Research problem: En compilation, comment concilier optimisations et préservation de propriétés de sécurité ?

Short description:

Les optimisations sont devenues un composant fondamental des compilateurs. Un compilateur moderne met en oeuvre des centaines d'optimisations, dont certaines sont génériques (s'appliquent à un programme indépendamment du processeur cible) et d'autres spécifiques au processeur (tentent d'exploiter au mieux les caractéristiques précises du processeur). Contre-intuitivement, une optimisation ne produit en aucun cas un programme *optimal*[1], mais elle améliore (avec une très forte probabilité) ses performances (le plus souvent temps d'exécution, mais on peut aussi "optimiser" la taille ou la quantité de mémoire, la consommation…)

Fondamentalement, une optimisation est une transformation appliquée au programme (strictement, elle produit donc un programme différent) qui améliore ses performances tout en préservant son *comportement observable* (en gros, son résultat).

Depuis quelques années, il est devenu évident que la recherche de performance ne devrait plus être le critère exclusif. En particulier, on veut de plus en plus du code performant mais aussi *sécurisé*. Du code sécurisé peut contenir des éléments fonctionnellement inutiles (leur présence ne change pas le comportement observable du programme). Par exemple, des tests de conditions peuvent être dupliqués (pour résister à des attaques par injection de faute[2]), des données sensibles qui ne seront plus utilisées peuvent être écrasées en mémoire (pour prévenir leur lecture par un attaquant suite à une vulnérabilité logicielle) ou le programme peut être écrit de manière à qu'à aucun moment un branchement ne dépende de la valeur d'une variable sensible (pour empêcher une attaque par canal caché temporel[3]).

Dans tous ces exemples, on a besoin de préserver des aspects *non fonctionnels* du programme, c'est à dire qui ne contribuent pas à son comportement observable. Les optimisations, ne garantissant la préservation que du comportement observable, peuvent altérer ces aspects.

Reference(s):

[1] The Correctness-Security Gap in Compiler Optimization. Vijay D'Silva, Mathias Payer, Dawn Song. 2015 IEEE Security and Privacy Workshops. http://dx.doi.org/10.1109/spw.2015.33

[2] What You Get is What You C: Controlling Side Effects in Mainstream C Compilers. Laurent Simon, David Chisnall, Ross Anderson. Euro S&P 2018

1 Des arguments théoriques montrent que c'est en général impossible… dommage !

2 Une attaque physique consistant à perturber le fonctionnement du matériel (en agissant sur un élément physique comme la tension d'alimentation, le signal d'horloge, l'environnement électromagnétique…) de manière à fausser le résultat d'une opération précise (ou pas) dans le programme.

3 Un canal caché temporel permet, en mesurant précisément le temps d'exécution d'un programme, de déterminer des éléments de son état interne qui ne sont pas fonctionnellement accessibles (par exemple des clés secrètes manipulées par le programme).

PLANARF: PLANar Antenna and RF Rectification

Supervisor: Darine Kaddour

Supervisor's email: Darine.Kaddour@grenoble-inp.fr

Research problem: What are the main design approaches and performance limitations of planar rectennas operating at 2.45 GHz for RF energy harvesting?

Short description: The aim of this work is to conduct a state-of-the-art review of planar rectennas operating at 2.45 GHz. The study will focus on four main aspects:

- Antenna: type, dimensions, gain, bandwidth and miniaturization techniques.
- Impedance matching: matching techniques between the antenna and the rectifier, such as LC networks, stubs and microstrip lines.
- Rectifier: main rectifier topologies, diode technologies and operation at low RF power levels.
- Performance: RF input power, DC voltage and power, RF-to-DC efficiency, load and overall dimensions.

The selected papers will be compared in a summary table in order to identify the main design trends, performance levels and trade-offs of planar rectennas at 2.45 GHz.

Students need to select recent scientific papers on planar rectennas and RF energy harvesting at 2.45 GHz, with priority given to peer-reviewed journal and conference papers.

References:

[1] H. Sun, Y.-X. Guo, M. He, and Z. Zhong, "Design of a high-efficiency 2.45-GHz rectenna for low-input-power energy harvesting," IEEE Antennas and Wireless Propagation Letters, vol. 11, pp. 929–932, 2012.

[2] M. Koohestani, J. Tissier, and M. Latrach, "A miniaturized printed rectenna for wireless RF energy harvesting around 2.45 GHz," AEU - International Journal of Electronics and Communications, vol. 127, Art. no. 153478, 2020.

Qcar_Control: Model Predictive Control and Differential Flatness-Based Approaches for the QCar: Control and Collision Avoidance Strategies

Supervisor: Ionela Prodan (ionela.prodan@lcis.grenoble-inp.fr)

Research problem: How have Model Predictive Control (MPC) and differential flatness-based control methods been applied to the QCar for trajectory tracking and collision avoidance? What dynamic models are typically used, are these models differentially flat? How is MPC formulated to handle constraints and obstacle avoidance in real-time operation?



Short description: This project aims to perform a state-of-the-art literature review on control strategies applied to the QCar (Quanser Connected Autonomous Vehicle Research platform), with emphasis on Model Predictive Control and differential flatness. Students will analyze how these approaches are used for: i) trajectory tracking under dynamic constraints, ii) path planning and real-time collision avoidance; iii) integration with perception and sensor fusion frameworks (e.g., LiDAR, camera). The review should focus on identifying:

1. The dynamic models used to describe vehicle motion (kinematic or dynamic bicycle models, flat or non-flat formulations);
2. Whether the chosen models are differentially flat and what flat outputs are used;
3. How MPC formulations incorporate system constraints (actuator limits, obstacle avoidance, road boundaries);
4. How collision avoidance is achieved (e.g., through mixed-integer constraints, soft constraints, potential fields or trajectory re-optimization).

Expected outcomes: i) literature review (at least 5 publications from the last 3 years) of existing QCar-based experimental and simulation studies on MPC and flatness-based control. ii) a comparative table describing which vehicle models were used, their flatness properties, and how MPC was applied for constrained control and collision avoidance.

References:

[1] Quanser. *QCar Overview and Specifications.* https://www.quanser.com/products/qcar/
[2] Rawlings, J. B., Mayne, D. Q., & Diehl, M. (2017). *Model Predictive Control: Theory, Computation, and Design.* Nob Hill Publishing.
[3] Fliess, M., Lévine, J., Martin, P., & Rouchon, P. (1995). *Flatness and Defect of Nonlinear Systems: Introductory Theory and Examples.* International Journal of Control.
[4] Wang, Wei, et al. "Autonomous vehicle trajectory following with flatness-based model predictive control and real-time iteration. Journal of Systems and Control Engineering (2025): 09596518251322238.
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RECONF-DET: Reconfigurable attack detectors in integrated circuits

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Research problem: How can on-chip detectors of physical attacks be designed to be configurable or reconfigurable, so that their sensitivity can be calibrated after fabrication and adapted at run time to the operating conditions and to the security needs of the circuit?

Short description: Integrated circuits embed digital sensors to detect physical fault injection attacks such as clock or voltage glitches, electromagnetic pulses or laser shots. Most of these detectors rely on fixed delay chains tuned at design time. They then suffer from process, voltage and temperature variations, from aging, and from voltage and frequency scaling, which cause false alarms or missed attacks. Reconfigurable detectors address these limits with programmable delay lines, adjustable thresholds, calibration loops and run-time activation. A well-known industrial example is the Tunable Replica Circuit integrated by Intel in its processors. The objective of this work is to review and classify reconfigurable attack detectors in integrated circuits (ASIC and FPGA). The study will compare: i) the detector architectures and the attacks they cover; ii) the reconfigurable parameters and the time at which they are set (design, test, boot, run time); iii) the calibration methods, on-chip or off-chip; iv) the reported results in terms of detection rate, false alarms, area and power.

Reference(s):

[1] C. Deshpande, B. Yuce, N. F. Ghalaty, D. Ganta, P. Schaumont and L. Nazhandali, "A Configurable and Lightweight Timing Monitor for Fault Attack Detection," *2016 IEEE Computer Society Annual Symposium on VLSI (ISVLSI)*, Pittsburgh, PA, USA, 2016, pp. 461-466, doi: 10.1109/ISVLSI.2016.123.

[2] Z. Chen, K. Yang and K. Ren, "CDS: An Anti-Aging Calibratable Digital Sensor for Detecting Multiple Types of Fault Injection Attacks," *2024 61st ACM/IEEE Design Automation Conference (DAC)*, San Francisco, CA, USA, 2024, pp. 1-6.

[3] M. Li, P. X. Huang, J. Park, S. K. Mathew, V. De and M. Seok, "GUARD: A Fully-Digital TDC-Based Clock and Voltage Glitch Detector with On-Demand Protection in a 28nm CMOS," *2025 Symposium on VLSI Technology and Circuits (VLSI Technology and Circuits)*, Kyoto, Japan, 2025, pp. 1-3, doi: 10.23919/VLSITechnologyandCir65189.2025.11075120.

SML: Similarity and Metric Learning

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Research problem: What are the different (frugal) methods for similarity/metric learning?

Short description:

Supervised Machine Learning applications generally focused on regression and classification tasks [1]. Particularly, a task of regression predicts a continuous output from input features, while a task of classification predicts a discrete output, called class, from input features.

Another related task, called similarity learning or metric learning, consists to learn a similarity function between two inputs (images, sounds, …), measuring the (semantic) proximity between those two objects [1]. The applications of this task can be for information retrieval, that consist of finding entries in a database that is most similar to the input. Particularly, it can be used for facial recognition, specific emitter identification, or for text generation. One of the first work on this task was for signature verification using siamese network by Y. Le Cun and his collaborators [2].

The study will identify the different methods proposed in the literature of similarity/metric learning. Particularly, after identifying the principal methods, the students can focus on frugal methods, that require low inter-class variability (few classes), as well as low intra-class variability (few examples per class).

Reference(s):

[1] I. Goodfellow, Y. Bengio, A. Courville – Deep Learning, The MIT Press, 2016

[2] J. Bromley, I. Guyon, Y. Le Cun, E. Säckinger and R. Shah, Signature Verification using a "Siamese" Time Delay Neural Network, NIPS, 1993

TDCFID: Time to Digital Converter for Fault Injection Detection

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Research problem: How can we use TDC (Time to Digital Converter), designed for very high precision physics particle and particle accelerator, in the context of hardware security ?

Short description: The objective of this work is to conduct a state-of-the-art study on the design and use of TDC in the context of hardware security. The study will identify and classify existing TDC architecture for both use in particle accelerator and for hardware security. Students will analyze methodologies and benchmarks proposed in the literature, and give an overview of existing works with advantages and constraints of each design.

Reference(s):

[1] M. R. Muttaki, M. H. Rahman, A. Kulkarni, M. Tehranipoor and F. Farahmandi, "FTC: A Universal Framework for Fault-Injection Attack Detection and Prevention," in IEEE Transactions on Very Large Scale Integration (VLSI) Systems, July 2024, doi: 10.1109/TVLSI.2024.3384531

[2] Henzler, S. "Time-to-Digital Converters." 2010, Springer Series in Advanced Microelectronics, doi: 10.1007/978-90-481-8628-0_2